



Coordination of forest management through market and political institutions☆☆☆

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ABSTRACT

Most forests worldwide are influenced or even dominated by political institutions governed by multilateral agreements. Forestry economic theory is yet mainly concentrated on bilateral exchanges within market institutions. This paper shows how the harvest and regeneration of trees as the objectively observable outcome of forest management vary with the institutional framework. Existing coordination theorems of market institutions are derived from a game theoretic interaction model which allows to extend the institutional setting towards political coordination of forest management. The analysis reveals that privately optimal harvest ages are restricted to fully internalized interdependence settings while external effects under collective ownership cannot be internalized via implicit coordination or exchanges of individual property rights. The derivation of an intertemporal model of public forestry goods provides empirically testable hypotheses on the evolution of forests within political institutional settings.

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1. Problem

The value people assign to forests is determined by the way trees are combined on the area. Different forest structures promise different flows of goods and services from the trees. The structure of the forest, conversely, is defined by the times of harvest and regeneration of the corresponding trees. Given the physical growing conditions of the relevant site, any forest structure is the consequence of a particular combination of harvests and regenerations of trees (including the omission to harvest or regenerate other trees). In order to explain and predict the evolution of forests as governed by the values people assign to them, it is thus necessary to study the emergence of the times of harvest and regeneration of interdependently growing trees.

Harvests and regenerations of trees are individual actions. Individuals, however, are not acting in isolation. Their actions are restricted by the actions of all other members of the relevant society, inclusive of the institutions, or rules, evolved out of past interactions. Self-sustaining and -enforcing action patterns as selected by the principles of evolutionary processes governing the social interaction nexus vary with the institutional framework. The diversity of forest structures,

and of the conflicts arising thereof, is thus bound to the diversity of institutional arrangements within which forest users interact.

The impact of different institutional settings on the harvest and regeneration of trees in forests yet remains a rather unresolved problem area. The main body of forest economic theory is focused on analyses within the specific institutional framework of effective market exchanges. Markets are the networks of relationships that emerge out of the bilateral trading process between people (Buchanan, 1964). The concentration on effective market institutions, however, might have been less motivated by observations, as many, if not most forests worldwide are influenced or even dominated by non-market institutions (cf. FAO, 2010), but possibly more by the analytical convenience or normative appeal of the consequences imposed by the commitment to this methodology.

Studies on the influence of institutional settings on the harvest ages of trees have been concerned with the ineffectiveness of markets (e.g. Tahvonen et al., 2001). Amacher et al. (2004) focus on non-internalized effects between forest owners within market institutions. The influence of personal characteristics of forest owners on harvest ages are analyzed by Tahvonen and Salo (1999). Other studies focus on the impact of political institutions on harvest decisions based on social welfare functions (e.g. Koskela and Ollikainen, 1999, 2001) or governmental preferences (Amacher, 1999). These analyses, however, imply the existence of rational and benevolent governments which conflict with the assumptions governing human actions in the market institutions employed. General resource economics, often with reference to fishery economics, provides a richer literature on institutional arrangements (e.g. Caputo and Lueck, 2003). The employment of density-dependent population models (based on the contribution by Schaefer,

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1954), however, are often only inappropriately applicable to forests as propositions concerning the forest structure are severely limited.

This paper aims at contributing to institutional diversity in forest economic theory by addressing a fundamental question: how do political institutions influence forest structures? What kind of management can be expected to evolve in forests which are governed by political institutions as contrasted with market institutions? Answers are proposed by resorting to a fundamental model of interaction processes regarding forest management based on a strict methodological individualism. Within a game theoretic framework, existing coordination theorems of market institutions – in particular, the Faustmann-Pressler-Ohlin (Johansson and Löfgren, 1985) and the associated Hartman theorem (Hartman, 1976) – are deduced from a property rights perspective which allows to address the fundamental differences to and consequences for politically coordinated forests. The derivation of an intertemporal model of public forestry goods eventually provides empirically testable hypotheses on the evolution of forests within a political institutional setting.

2. Methodology and method

Methodologically, the analysis is placed within the sub-constitutional stage of economic theory in which individuals interact within a given institutional setting (Buchanan, 1975). The rules or institutions, as emerging within a constitutional stage, define the ways and margins in which resources can be disposed of by the individuals. As the basic unit of discretionary power, a property right shall be defined. Property rights are the claims to the granting of services or to the allocation of goods to which the owner is entitled by the recognition of all other members of the society. They define the way that rents accruing from the use and regulation of goods and services, including natural resources and forests, are distributed among the corresponding owners. In contrast to a formal title assigned by legislative authorities, the mutual, unanimous recognition of property rights necessarily entails their enforcement.¹

Within the sub-constitutional stage, the analysis distinguishes between the two basic settings of market and political institutions (Buchanan, 1968). Market institutions are characterized by the bilateral exchange of private property rights held exclusively by a single individual. Political institutions, by contrast, are defined as the rules for collectively held property rights which are owned by more than one individual. Since the disposal over collective property rights requires the consent of the rights holders, exchange within political institutions is effected by means of multilateral agreements.

Both market and political institutions might be distinguished within the external and the extended order of moral conduct (Deegen, 2013). The following analysis, though, is exclusively confined to the extended order of impersonal interactions (Hayek, 1979; Smith, 2007).² Anonymous exchange within a large group setting is guided by the observance of abstract rules and signals. Exchange within the external order, on the other hand, whether via politics or markets, imposes discretionary strategic alliances on the analysis whose outcome depends on the personality of the individuals involved. Although the logic of group interactions is independent of the group size, discretionary strategic behavior in small groups leaves ample scope for situations which are not, or only inappropriately, manageable with the economic axiom of rational adaption of individuals to dilemmata situations.³

Methodically, the analysis is conducted within a partial equilibrium setting. Endogenous adaptations are restricted to a partial interaction nexus embedded within a general nexus of interaction determined by the relevant society. Exchange between the interaction nexuses takes place via effective market institutions. This construction conveniently allows to perform the analysis within commodity space as the utility setting can be normalized by exchanges with the general interaction nexus independent of personal idiosyncrasies (for a transformation of the commodity setting into utility space cf. Amacher et al., 2009, 41f.).

The partial equilibrium setting moreover allows to abstract from the observable, intensive intermixture of institutional settings. Market and political institutions can appear on either side of the demand and supply side of exchange processes. Private goods and services traded via markets can be supplied collectively, while collective demand organized via political processes can be supplied privately. With the partial equilibrium setting, supply and demand, just as production and consumption, can be heuristically combined such that each holder of property rights to forests within the interaction sub-nexus supplies and demands trees simultaneously.

Finally, the large group setting is handled with the help of a small group framework. The relevant characteristics of large group interactions are retained by commitment to symmetric equilibria and neglect of strategic behavior. Parametrical responses on the part of the individuals analyzed guarantee the concentration of the study on anonymous interactions within the extended order. Any personal references in the analysis are only due to this heuristic.

3. Model

Given effective markets for the in- and outputs of forest management within the general interaction nexus, the choice set of forest land owners determining their land value is restricted to the harvest ages of the trees,⁴ T_i . Interacting in a competitive environment, forest owners are selected by principles requiring the maximization of the land value under the prevailing price set. The emerging outcome is defined as the Faustmann harvest age,⁵

$$T_i^F := \arg \max_{T_i \in \mathbb{N}} V^i = v^i(T_i) e^{rT_i} - c_i, \quad (1)$$

where V^i is the land value, v^i is the tree value, c_i are potential regeneration expenses and r is the rate of interest. The corresponding first-order condition demands that

$$\frac{dV^i(T_i^F)}{dT_i} = v_{T_i}^i(T_i) e^{-rT_i} - r v^i(T_i) e^{-rT_i} = 0 \quad (2)$$

where subscripts to functions indicate derivatives. As long as the second-order derivative,

$$\frac{d^2 V^i}{dT_i^2} = v_{T_i T_i}^i(T_i) e^{-rT_i} - r v_{T_i}^i(T_i) e^{-rT_i}, \quad (3)$$

is negative throughout the considered range, a possible solution to the first-order condition in [2] is unique.

Interactions within the sub-nexus emerge according to mutually dependent modes of action captured by a non-cooperative game $G = \langle N, (A_i), (U_i) \rangle$. The player set shall be given by $N = \{k, -k\}$, where $-k$

¹ Vice versa, though, legal enforcement mechanisms may be part of the recognition process. Unenforceable rights, on the other hand, are not property rights in this setting.

² “Political institution” might be an appropriate term for collectively held rights in the extended order, while collective institutions might then refer to a generic term for collective rights independent of the moral order.

³ Economics, or understood as catallactics (cf. Deegen and Hostettler, 2014), can be defined as the science of voluntary exchanges within complex social networks. Following this conception, economic theory might only be applicable to large or small group settings within the extended moral order. The external order, by contrast, is then a mixed field of application together with the science of the internal order of the mind or psychology.

⁴ There can be one or several harvest ages for n trees or stands with $T_i = (T_i^1, \dots, T_i^n)$. In the one-period setting, time and age are synchronized necessarily.

⁵ The infinitely periodical nature of the Faustmann model is fully acknowledged. The term “Faustmann” is employed as a representative of market institutions. In order to distinguish the periodical setting of the Faustmann rotation age from the one-period case considered here, the term Faustmann harvest age is employed. The Faustmann harvest age represents the special case of the Faustmann rotation age with zero land value. It is vindicated by the emergence of the genuine Faustmann model below.

might also corresponds to “all other players” with $-k = (1, \dots, k-1, k+1, \dots, n)$. (A_i) is the action space, where the individual action set, A_i , for each player $i \in N$ is given by the harvest age $\{T_i\}$. The payoff space (U_i) defines a payoff for each player given by the land values defined in Eq. (1). The relevance of the game stems from the mutual impacts of the harvest ages on the tree values, i.e.

$$v^i = v^i(T_k, T_{-k}). \quad (4)$$

Interaction equilibria in game G , characterized by the lack of incentives for deviations from the status quo, can be described by Nash equilibria defined as

$$V^i(T_k^N, T_{-k}^N) = \max_{T_i > 0} V^i(T_i, T_{-i}^N) \quad \forall i \in N \quad (5)$$

where T_i^N are the Nash harvest ages. Following the maximization paradigm, Nash equilibria can also be defined in terms of best response correspondences describing individual adaptations to choices made by other players. If b^i is the best response correspondence of player i , then it must hold at a Nash equilibrium that

$$b^k(T_{-k}^N) = b^{-k}(T_k^N). \quad (6)$$

The best response correspondences are implicit in the individual first-order conditions, cf. Eq. (2), on grounds of

$$\left. \frac{\partial V^i[T_i, b^{-i}(T_i)]}{\partial T_i} \right|_{(T_i^N, T_{-i}^N)} = v_{T_i}^i[T_i, b^{-i}(T_i)] e^{-rT_i} - r v^i[T_i, b^{-i}(T_i)] e^{-rT_i} = 0. \quad (7)$$

If v^i is twice continuously differentiable and the second-order derivatives

$$\frac{d^2 V^i[T_i, b^{-i}(T_i)]}{dT_i^2} = v_{T_i T_i}^i[T_i, b^{-i}(T_i)] e^{-rT_i} - r v_{T_i}^i[T_i, b^{-i}(T_i)] e^{-rT_i} \quad (8)$$

together with the corresponding Hessian determinant

$$|H_i| \equiv \begin{vmatrix} \frac{\partial^2 V^i}{\partial T_i^2} & \frac{\partial^2 V^i}{\partial T_i \partial T_{-i}} \\ \frac{\partial^2 V^i}{\partial T_{-i} \partial T_i} & \frac{\partial^2 V^i}{\partial T_{-i}^2} \end{vmatrix} \quad (9)$$

are non-zero, the implicit function holds at points satisfying Eq. (7). Since b^{-i} is implicitly formulated, it is only possible to infer on the slope of the best response correspondences. Implicit differentiation of Eq. (7) yields

$$\frac{db^{-i}(T_i)}{dT_i} = - \frac{\frac{\partial^2 V^i}{\partial T_i^2}}{\frac{\partial^2 V^i}{\partial T_i \partial T_{-i}}}. \quad (10)$$

As long as the second-order condition holds, the sign of Eq. (10) depends on the denominator. With the partial-cross derivative given by

$$\frac{\partial^2 V^i(T_i, T_{-i})}{\partial T_i \partial T_{-i}} = v_{T_i T_{-i}}^i(T_i, T_{-i}) e^{-rT_i} - r v_{T_{-i}}^i(T_i, T_{-i}) e^{-rT_i}, \quad (11)$$

the sign of Eq. (10) thus varies as

$$\frac{db^{-i}(T_i)}{dT_i} \lessgtr 0 \text{ if } \frac{v_{T_i T_{-i}}^i}{v_{T_{-i}}^i} \gtrless r \quad (v_{T_{-i}}^i \neq 0). \quad (12)$$

In contrast to temporal settings referring to production functions, the growth function of the tree value includes stock effects through which the best response correspondences are dependent on the interest

on value increments. As a consequence, situations may arise in which the slope of the best response correspondences share the same sign with the partial-cross derivatives. Eq. (12) corresponds to the concept of spatial and temporal dependence developed by Koskela and Ollikainen (2001) and Amacher et al. (2004) without the restriction to flow variables.

Due to the assumption of continuous payoff functions, there is a pure-strategy Nash equilibrium necessarily (Debreu, 1952) to which the analysis is restricted. Eventually, given the monotony of the best response correspondences implied by the fulfillment of the second-order condition, the Nash equilibrium is unique.

4. Analysis

4.1. Market exchange

The problem situation of the interaction game formulated above arises from the mutual dependence of the tree values on the actions of the forest owners. Though both are acting within a market institutional setting through which inputs and outputs to the forest management process are exchanged, the interdependencies of the tree values are not internalized. The selection of a harvest age not only determines the own, but also the tree values of the other individual(s).

As an example, consider the case of ecological competition between two adjacently growing trees (or stands), each held by one of the forest owners defined above.⁶ Due to their proximity, the trees shall share equivalent growth functions q^i . Both owners are selling the tree product on a competitive market at a price p . Hence, $v^i = p q^i(T_i, T_{-i})$ with $q^i > 0$, $q_{T_i}^i > 0$, $q_{T_{-i}}^i < 0$ and where $q_{T_{-i}}^i < 0$ and $q_{T_i T_{-i}}^i > 0$ for $T_i > T_{-i}$ while $q_{T_{-i}}^i = q_{T_i T_{-i}}^i = 0$ for $T_i \leq T_{-i}$.

The resulting best response correspondences are sketched in Fig. 1. The Faustmann harvest age, T_i^F , from condition (2) evolves in the absence of competition where the best response correspondences cut the axes. With competition, trees are mutually impeding their growth while growing simultaneously. When each owner takes the action(s) of the other(s) as given, they are equivalent to growing conditions which, if lasting throughout the relevant period, lead to harvest ages of T_i^F . Through the mutual interaction, though, both owners will harvest their trees at the point marked by the intersection of the best response correspondences. According to Eq. (6), this is the Nash equilibrium since no owner has an incentive to deviate by harvesting his tree earlier or later. In the face of negative interdependencies, the Nash equilibrium is shorter than the Faustmann harvest age and longer than an environmentally imposed harvest age T_i^E .

Table 1 illustrates the attainment of the different equilibria by means of a calculation example. The applied timber growth function for Loblolly pine (*Pinus taeda* L., converted into m³ and ha) is taken from Chang (1984), which has been extended in order to allow for competition effects (cf. Eq. (A.1) in the Appendix A). According to condition (2), the Faustmann harvest age (F) is determined by the equality of the rate of value growth with the rate of interest at an age of 20 years (column 4) yielding a land value of about \$997 ha⁻¹ (column 5). Due to the competition effects, though, the Faustmann harvest age and land value is not realizable. Instead, the stand will be harvested at the Nash equilibrium (N) when the best response correspondences are equally at an age of 18 years (column 6). The corresponding land value in column 7 is about 36% lower than the Faustmann land value.

The efficiency (and thus the enforceability) of the Nash equilibrium hinges on the availability of viable alternative institutional

⁶ Naturally, value interdependencies between trees may arise out of countless other sources.

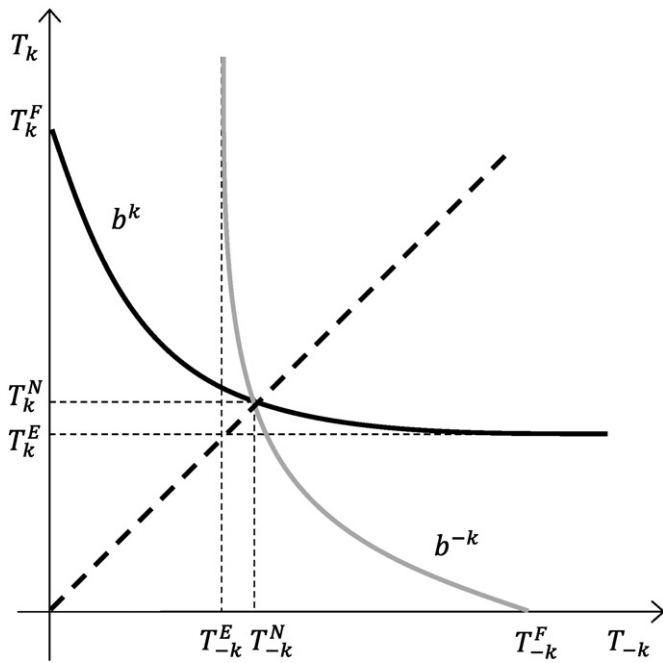


Fig. 1. Best response correspondences with negative interdependencies between privately held trees.

arrangements. Considering the joint maximization problem including both (or all) trees given by

$$\max_{T>0} V = \sum_{j \in N} [v^j(T) e^{-rT} - c_j] \quad (13)$$

where V is the total land value and $T := T_j = T_{-j} \forall j \in N = \{k, -k\}$ is the uniform harvest age, the corresponding first-order condition is given by

$$\frac{\partial V(T^+)}{\partial T} = \sum_{j \in N} v_t^j e^{-rT} - r \sum_{j \in N} v^j e^{-rT} = 0 \quad (14)$$

where T^+ denotes the optimal joint harvest ages. As long as the trees share equal value growth functions $v := v^j = v^{-j} \forall j \in N = \{k, -k\}$, the equilibrium under the specified competition pattern is determined by

$$nv_T - nrV = 0 \Leftrightarrow v_T = rv \quad (15)$$

which yields the same solution as Eq. (7) for all T . Hence, under the assumptions of equal growth and harvest ages, the Nash equilibrium is efficient to the effect that the forest owners cannot improve their situations through a coordinated approach or trade. In the calculation example in Table 1, the optimal joint harvest age is 18 years yielding a total land value of twice the single land value in column 7, or \$1283 ha⁻¹.

The absence of trading opportunities is the consequence of the equality assumption employed. The effective market exchange within the general interaction nexus allows to perform the partial analysis entirely within commodity space. Through this assumption, individual marginal utility is adjusted to price ratios such that the incentives to produce are equivalent for all participants. In combination with equivalent growing conditions, this assumption naturally leads to equivalent harvest ages within the symmetric game theoretical framework. Trade, though, can only be mutually beneficial in the presence of some inequalities which can be mutually exploited.

If the assumption of equal harvest ages is relaxed, various trading opportunities emerge. The overall land value can be increased if

negative interdependencies between the tree growth functions (e.g. due to ecological competition) can be internalized. In this situation, the previous harvest of one tree offers the opportunity to increase the value of the other tree (even if the trees are growing equivalently). If both owners coordinate forest management within a joint enterprise (at zero additional cost), the corresponding optimality condition derived from the maximization of the objective function,

$$\max_{T_k, T_{-k} > 0} V = \sum_{j \in N} [v^j(T_k, T_{-k}) e^{-rT_j} - c_j], \quad (16)$$

demands that⁷

$$\frac{\partial V(T_k^*, T_{-k}^*)}{\partial T_k} = v_{T_k}^k e^{-rT_k} - rv^k e^{-rT_k} + v_{T_k}^{-k} e^{-rT_{-k}} = 0 \quad (17)$$

$$\frac{\partial V(T_k^*, T_{-k}^*)}{\partial T_{-k}} = v_{T_{-k}}^{-k} e^{-rT_{-k}} - rv^{-k} e^{-rT_{-k}} + v_{T_{-k}}^k e^{-rT_k} = 0, \quad (18)$$

where T_i^* are the optimal harvest ages, to be satisfied simultaneously. Accordingly, and unless there are sufficiently positive interdependencies between the tree values offering incentives for simultaneous harvests (e.g. due to costly harvest operations), the trees will be harvested at different ages since the marginal impact of the trees on each other are different once a tree is harvested. In forestry terms, the forest is thinned.

The thinning case is illustrated in Fig. 1 by the points enclosed by the best response correspondences. In each of these cases, the harvest ages of the two trees diverge. However, with the restriction to equal harvest ages illustrated by the dashed diagonal, the non-cooperative Nash equilibrium remains as the only viable solution. In the calculation example in Table 1, the total land value can be increased by harvesting the trees (or stands, respectively) at different ages of 16 and 18 years (as marked by the asterisks in column 8).

Internalization of interdependent tree growth through thinning, however, is not enforceable via implicit coordination as in the case of equal harvest ages. The thinning solution does not qualify as a Nash equilibrium since thinning is unilaterally realizing a lower land value (cf. column 7 at age 16 in Table 1). The expected payoff of thinning is lower thus offering incentives to postpone the harvest. Internalization can yet be achieved by bilateral exchanges of property rights within market institutions as long as transaction costs are non-restrictive. Given a higher total land value, each of the forest owners considered can offer the other owner(s) a payment for the property rights to the land greater than the land value determined by the Nash equilibrium.⁸ In Table 1, each owner can increase his income by offering the other a payment of up to \$683 ha⁻¹ for his land rights, which is more than the expected Nash land value.

4.2. Open access

The interaction dependencies of the forest owners in the game above tend to disguise the institutional solution to a more fundamental interaction problem which has already been obtained. The effective market exchange within the entire interaction nexus rests on the assumption of recognized and enforceable rules underlying the exchange process. The existence of a rule set guaranteeing bilateral exchange opportunities influences the forest structure as the observable outcome of the interaction process significantly.

The market outcome above depends on the property rights assigned and recognized by all other participants within the considered interaction nexus. In particular, the indices to the investment parameters of

⁷ In general, T_{-k} might denote the harvest ages of all trees of a stand other than k .

⁸ Alternatively, only single property rights of the bundle defining full land ownership can be traded, e.g. harvest rights.

Table 1
Calculation example for negative interdependencies between privately held trees.

1	2	3	4	5	6	7	8
T_i	$q^i(T_i, 0)$	$q_{T_i}^i(T_i, 0)$	$q_{T_i}^i/q^i$	$V^i(T_i, 0)$	$b^i(T_{-i})$	$V^i(T_i, T_{-i}^N)$	$V(T_i, T_{-i}^*)$
[a]	[m ³ ha ⁻¹]	[m ³ ha ⁻¹ a ⁻¹]	[a ⁻¹]	[\$ ha ⁻¹]	[a]	[\$ ha ⁻¹]	[\$ ha ⁻¹]
5	3.90	5.34	1.370	−356.69	19.40	−359.91	289.6
6	11.76	10.45	0.889	−276.94	19.28	−287.86	362.4
7	24.70	15.31	0.620	−156.52	19.15	−181.53	470.2
8	42.03	19.15	0.456	OA − 9.85	19.03	OA − 55.31	598.6
9	62.61	21.81	0.348	147.26	18.90	76.06	732.6
10	85.30	23.42	0.275	302.18	18.78	201.44	861.0
11	109.17	24.21	0.222	446.39	18.67	313.82	976.6
12	133.53	24.41	0.183	574.91	18.55	409.57	1075.6
13	157.85	24.18	0.153	685.37	18.44	487.45	1156.6
14	181.79	23.66	0.130	777.19	18.33	547.73	1220.0
15	205.11	22.96	0.112	850.88	18.22	591.58	1266.6
16	227.67	22.15	0.097	907.62	18.12	620.63	*1298.2
17	249.39	21.28	0.085	948.93	18.01	636.70	1316.6
18	270.22	20.38	0.075	976.47	N 17.91	N 641.60	*1323.5
19	290.14	19.48	0.067	991.90	17.81	637.10	1320.7
20	309.18	18.59	F 0.060	F 996.84	17.71	624.78	1309.8
21	327.34	17.73	0.054	992.76	17.61	606.09	1292.2
22	344.66	16.91	0.049	981.05	17.52	582.30	1269.3
23	361.17	16.12	0.045	962.94	17.43	554.52	1242.2
24	376.91	15.37	0.041	939.51	17.34	523.69	1211.8
25	391.92	14.66	0.037	911.74	17.25	490.65	1178.9

$r = 0.06$ p.a.; $p = \$15$ m³; $c_k = \$400$ ha⁻¹; $m = 300$; $s = 80$.

the objective functions in Eq. (1) define the relevant property rights claims in the game considered. The right to take possession of the harvested tree is held by the correspondingly indexed individual to the tree value. In the same way, the person concerned holds the rights to his actions implied by the indexed harvest age (he is not a slave) and to the regeneration expenditures he might issue. The income stream arising from this bundle of rights determines the land value for the individual considered.

In the absence of property rights to the forest management process, the emerging forest structure is modified decisively. Without rule sets delimiting the individual spheres of discretionary power over forest uses, none of the potential forest users can be prevented from taking actions within the anarchical domain (at least at low cost). This free, unconditional use of natural resources including forests is typically described as “open access”⁹ (Clark, 2005, p. 24ff.).

Consider the case without property rights to the harvested trees. For the problem situation depicted above, the individual problem becomes to

$$\max_{T_i > 0} V^i = \sum_{j \in M} [v^j(T_k, T_{-k})e^{-rT_i} - c_j] \quad (19)$$

with $M = \{1, \dots, m\}$ as the set of trees accessible. Under these circumstances, each forest owner faces the opportunity to harvest trees. The missing personal index to the tree values implies the lack of property rights to the trees. In this case, the interdependencies between the tree values to the individuals take extreme values as the trees are (assumed to be) fully rival. If $-k$ harvests the trees, then k cannot harvest them and cannot receive any income.

Let T_i^0 denote the open access harvest age, the first-order condition to problem (19) is

$$\frac{\partial V^i(T_i^0, T_{-i})}{\partial T_i} = \sum_{j \in M} v_{T_i}^j e^{-rT_i} - r \sum_{j \in M} v^j e^{-rT_i} = 0 \quad (20)$$

⁹ In this study, open access refers to the lack of property rights. Hence, it may occur in varying degrees ranging from the absence of single property rights to full open access without any formal or informal rules of conduct.

which is equivalent to condition (15) for $T = T_i$ and $n = m$. Without (relevant) interdependencies, the equilibrium is thus described by the Faustmann harvest age. With mutually dependent tree values, on the other hand, the slope of the best response correspondences is determined by the partial cross derivative,

$$\frac{\partial^2 V^i(T_i, T_{-i})}{\partial T_i \partial T_{-i}} = \sum_{j \in M} v_{T_i T_{-i}}^j e^{-rT_i} - r \sum_{j \in M} v_{T_{-i}}^j e^{-rT_i}. \quad (21)$$

Since Eq. (21) is equivalent to the negative of the second-order derivative as T_i and T_{-i} describe oppositely directed actions, the slope of the best response correspondences in Eq. (10) is equal to 1 as long as the tree value is non-negative and the harvest ages are lower than the Faustmann harvest age.

The situation is illustrated in Fig. 2. Both forest users abstain from harvesting if the tree values are negative since the alternative not to harvest yields a higher income. Hence, the best response correspondences in Fig. 2 initiate at the cost-covering harvest ages of the trees (T_i^{cc}). Once the trees have crossed the cost-covering age, though, it is optimal to harvest the tree just before the other user(s) (illustrated by the jump of the best response correspondences after the cost-covering diameter; blank circles are not part of the correspondences). Eventually, the trees are not harvested beyond the Faustmann harvest age (thereby inducing axially parallel best response correspondences) since the land value decreases thereafter by virtue of Eq. (15).

As Fig. 2 reveals, the unique Nash equilibrium of the open access game with non-assigned property rights to harvested trees is characterized by the cost-covering diameter. Hence, tree and land values are zero. In the presence of regeneration cost, land values are even negative thus offering no incentives for investments at all. Only trees with zero regeneration cost (e.g. spontaneous natural regeneration) will emerge which will be harvested soon after cost coverage.

In Table 1, the open access outcome is illustrated in columns 5 and 7. If the compounded regeneration expenses are taken as harvest cost, the open access equilibrium (OA) is attained at a harvest age of 8 years, where the stand comprises a timber volume of 42 m³ ha⁻¹ and the land value is zero. Within the displayed range, the position of the cost-covering harvest age is constant over the harvest ages.

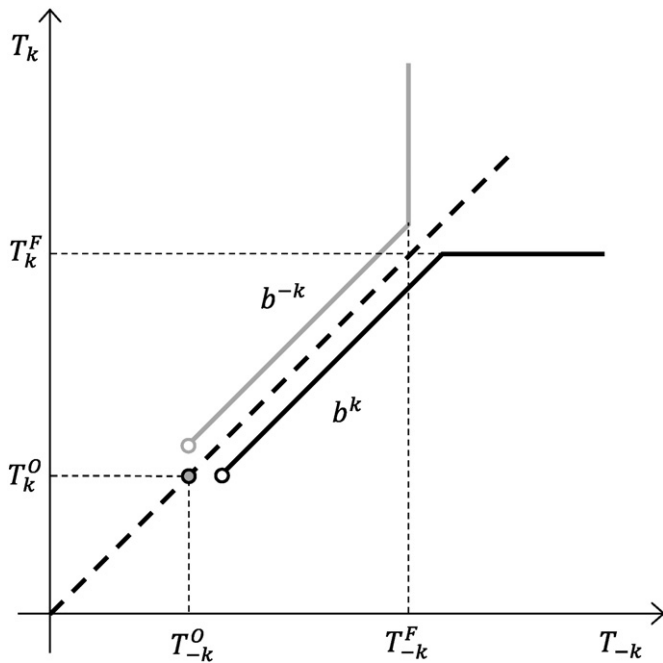


Fig. 2. Best response correspondences in the absence of property rights to the harvested trees.

The exploitation of the forest through selective harvests of cost-covering trees characterizes an open access dilemma¹⁰ in which no one gains. Although all forest users could benefit from higher harvest ages, missing property rights to the harvested trees do not allow for the expectation of future income.¹¹ Both (or all) forest users are trapped in a “prisoner’s dilemma” in which all have incentives to defect.

Compared to the open access outcome, both forest users are better off with property rights to the harvest of a tree as in Fig. 1 when competition between the trees allows for a crossing of the cost-covering diameter. The extreme interdependencies in the open access situation can thus be reduced (and often have been) by the mutual recognition of property rights to the trees.¹²

4.3. Public good

While the recognition of property rights is a collective process within a constitutional contract (Buchanan, 1975, p. 33ff.), the assigned property rights can be held privately or collectively. In contrast to private rights, collective property rights are held by at least two individuals. As a consequence, the exercise or omission of collective actions requires the consent of all rights holders. Whereas no one can be prevented from harvesting trees in the open access case considered above, collective ownership is exclusively bound to a specific group of individuals as rights holders. If collective rights are assigned to the harvested trees, the management of the forest

becomes a public good.¹³ All rights holders are forced to adjust to the same quantity or quality of the corresponding good or service through the institutional setting given by the constitutional arrangement.¹⁴

Consider the case where there is an additional value, w , associated with the trees to which usage both owners hold equal property rights: both must adjust to the same quantity/quality of that good (or service), which under these circumstances is typically characterized as public. The rights to the harvested trees, on the other hand, are privately held by each owner as in the situation depicted in Fig. 1. The individual problem is to

$$\max_{T_i > 0} W^i = v^i(T_i)e^{-rT_i} - c_i + w(T_k, T_{-k}). \quad (22)$$

where W^i is the corresponding land value. For convenience, w is assumed to be a present value flow variable. Recreation might serve as an example. The missing index to the public good implies that both individuals have no right to exclude the other(s) from visiting the forest for recreational activities.¹⁵

The first-order condition to problem (22) demands that

$$\frac{\partial W^i(T_i^H, T_{-i})}{\partial T_i} = v_i^i e^{-rT_i} - r v_i^i e^{-rT_i} + w_{T_i}(T_k, T_{-k}) = 0. \quad (23)$$

Without interdependencies, individuals choose the Hartman harvest age¹⁶ (Hartman, 1976), T_i^H , which includes the compromise between the single maxima of the different products (v^i and w) harvested simultaneously. With interdependencies, the sign of the best response correspondences is determined by virtue of

$$\frac{\partial^2 W^i(T_i^H, T_{-i})}{\partial T_i \partial T_{-i}} = w_{T_i T_{-i}}, \quad (24)$$

cf. Eq. (10). If the public good is characterized by a positive, concave value function, then $w_{T_i T_{-i}}$ and the slopes of the best response correspondences are negative.

A possible problem situation is displayed in Fig. 3, which rests on the assumption that the highest quantity/quality of the public good is achieved if one owner harvests his trees at the Hartman harvest age. In the latter case, the maxima of the private and the public good for the other owner coincide such that the Faustmann harvest age emerges by virtue of condition (23). The symmetric Nash equilibrium with the public good, denoted by T_i^P , is located between the Faustmann and Hartman harvest ages as indicated by the intersections in Fig. 3. Each owner gains some (e.g. recreational) value through the production of the other owner. Neither of the owners is forced to produce the entire good for himself and both must pay for it by means of deviances from the Faustmann harvest ages.

Table 2 extends the calculation example in Table 1 by a public good setting. Next to private and independent (i.e. $T_{-i} = 0$ in Eq. (A.1) in the Appendix A) timber production, the trees generate income in terms of a present value flow w . The corresponding concave function is adapted from Swallow et al. (1990) and adjusted to include symmetric harvest age dependencies, cf. Eq. (A.2) in the Appendix A. The parameters

¹³ “Public”, in the sense used in this paper, refers to the whole relevant society including every member.

¹⁴ According to this approach, the non-rivalry usually defining public goods does not stem from the (physical) characteristics of the good, but from the property rights assigned to them (although the good’s characteristics may be an important, yet not the only determinant for the assignment of property rights on the constitutional level).

¹⁵ This problem situation characterizes a public good version of the Hartman model (Hartman, 1976). In contrast to the latter, forest owners must adjust to the same quantity/quality of the additional value. In the Hartman model, on the other hand, forest owners adapt to the simultaneous production of privately held goods.

¹⁶ As with the Faustmann harvest age, the difference to the original, rotational Hartman model is fully acknowledged by the denotation as Hartman harvest ages, cf. footnote 6.

¹⁰ Cf. the “tragedy of the commons” as coined by Hardin (1968), which is effectively an open access tragedy (Bromley and Cernea, 1989). “Commons” might be more suitably analyzed within an external order setting (cf. Deegen, 2013).

¹¹ The open access outcome also emerges in the presence of a zero-land-value condition underlying the individual problem in Eq. (19). This condition implies the equality between total benefits and costs (Gordon, 1954), where all rents dissipate due to the free access.

¹² Cf. Buchanan (1975) for the relevance of property rights for the movement out of the Hobbesian state of nature and for the development of humans to persons.

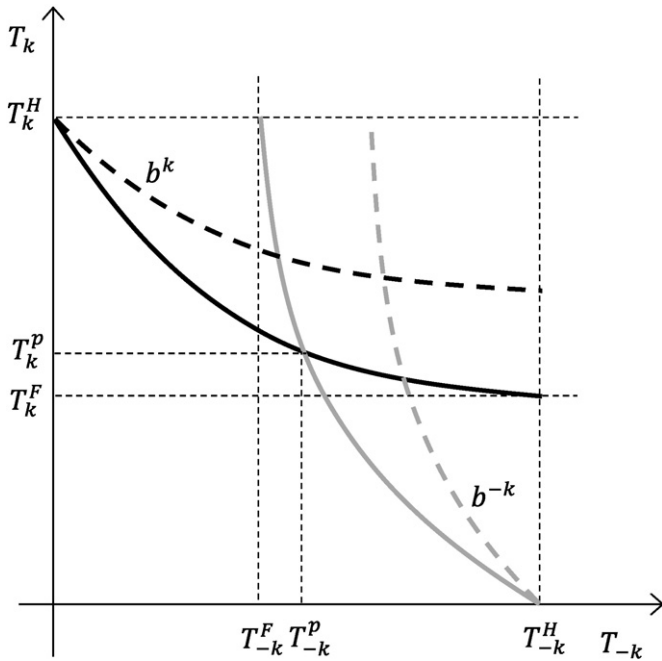


Fig. 3. Best response correspondences in the presence of a public good.

have been selected in order to illustrate the public good problem within the range considered. Column 3 reveals that the optimal Hartman harvest age (H) is attained at an age of 28 years, or 8 years after the Faustmann harvest age (F) taken from Table 1. Following the best response correspondence in column 4, the (public good) Nash equilibrium (N) emerges 6 years before the Hartman age yielding a land value of about $\$3217 \text{ ha}^{-1}$ (column 5) or $\$587 \text{ ha}^{-1}$ more than the Hartman solution. If the other owner restricts his action set to the Hartman harvest ages, the forest owner's best response is to harvest his tree(s) at the age of 21 years (I) thus increasing his land value relative to the Nash equilibrium and the Hartman age.¹⁷

The individual responses to the public good situation in a non-cooperative game setting may diverge from the outcome of a cooperative solution. Considering a joint enterprise with equal harvest ages for the trees, the problem becomes to

$$\max_{T>0} W = \sum_{j \in N} [v^j(T)e^{-rT} - c_j] + w(T) \quad (25)$$

with $T := T_j = T_{-j} \forall j \in N = \{k, -k\}$ and where W is the total land value. If T^j denotes the optimal joint harvest age, the necessary first-order condition is

$$\frac{\partial W(T^j)}{\partial T} = \sum_{j \in N} v_T^j e^{-rT} - r \sum_{j \in N} v^j e^{-rT} + w_T = 0. \quad (26)$$

For n homogenous trees with $v := v^j = v^{-j} \forall j \in N = \{k, -k\}$, the first-order condition thus demands that

$$v_T + \frac{w_T}{n} e^{rT} = rv. \quad (27)$$

In contrast to the individual adjustments with separated property rights to the trees in Eq. (23), joint maximization directly accounts for

¹⁷ If the valuation of the public good enters the individual utility directly, income effects become relevant such that the analysis cannot be conducted in commodity space (Tahvonen and Salo, 1999; Amacher et al., 2009, p. 71f.). Income effects can also gain importance in relation to the group size (Chamberlin, 1974).

the effects of the trees on the public good. Since all trees may contribute to the public good, only the share of each marginal evaluation enters the first-order condition.¹⁸ Compared to the individual Hartman harvest ages, the collective multiple-product harvest age is lower, given a positive marginal evaluation of the collective good, but higher than the Faustmann harvest age.

Compared to the Nash equilibrium, on the other hand, the cooperative outcome might diverge in either direction since $w_{T_k}(T_k, T_{-k}^N)$ can be equal to, greater or less than w_T/n . Though only little might be supplied by each in the Nash equilibrium, the sum of all contributions might well exceed the amount/quality under joint maximization (cf. Fig. 3 for two diverging scenarios). In contrast to the situation with exclusively private property right, however, a coordinated approach through joint maximization yields higher (or at least equal) land values than the Nash equilibrium.

In the example in Table 2, the optimal joint harvest ages (J) are located at an age of 24 years as revealed by column 7. Since the production interdependencies are internalized by virtue of condition (27), the optimal harvest ages are longer (and total supply of the public good larger) than in the Nash equilibrium. Compared to the latter, though, the total land value increases only slightly due to the low number of the participants in the example.

As with private property rights, unequal harvest ages likewise allow for the specialization of the production process. Considering the joint approach to the public good production with potentially unequal harvest ages,

$$\max_{T_k, T_{-k} \geq 0} W = \sum_{j \in N} [v^j(T_j)e^{-rT_j} - c_j] + w(T_k, T_{-k}) \quad (28)$$

together with the corresponding first-order condition,

$$\frac{\partial W(T_k^{**}, T_{-k}^{**})}{\partial T_k} = v_{T_k}^k e^{-rT_k} - rv^k e^{-rT_k} + w_{T_k} = 0 \quad (29)$$

$$\frac{\partial W(T_k^{**}, T_{-k}^{**})}{\partial T_{-k}} = v_{T_{-k}}^{-k} e^{-rT_{-k}} - rv^{-k} e^{-rT_{-k}} + w_{T_{-k}} = 0, \quad (30)$$

with T_i^{**} as the optimal harvest ages, the owners have the opportunity to concentrate the production of the public and private good on different trees. For instance, one tree might be harvested at the Hartman while the other is harvested at the Faustmann age by virtue of

$$v_{T_k}^k e^{-rT_k} + w_{T_k} = rv^k e^{-rT_k} \quad (31)$$

$$v_{T_{-k}}^{-k} e^{-rT_{-k}} = rv^{-k} e^{-rT_{-k}}. \quad (32)$$

Considered separately, both boundary solutions might thus be observable although each only emerges within the system of both products.

In Table 2, a joint maximization of the land value with variable harvest ages leads to the harvest of the stands (or trees) at the ages of 22 and 26 years in column 8. Accordingly, partial (yet not total) specialization of the stands on the two products appears to be efficient in the example. The gain in the land value, on the other hand, is only low (due to the low number of participants in the example).

In contrast to exclusively private property rights, however, unequal harvest ages of trees owned by different individuals sharing a public value are neither enforceable through exchanges of property rights nor via (implicit) coordination. While market institutions are characterized by the bilateral exchanges of property rights, trade in political

¹⁸ The outcome in Eq. (27) corresponds to the Samuelson condition for public goods in utility space (Samuelson, 1954).

Table 2
Calculation example with a public forest good.

1	2	3	4	5	6	7	8
T_i	$w(T_i, 0)$	$W^d(T_i, 0)$	$b^i(T_{-i})$	$W^d(T_i, T_{-i}^p)$	$W^d(T_i, T_{-i}^H)$	$W(T)$	$W(T_i, T_{-i}^*)$
[years]	[\$ ha ⁻¹]	[\$ ha ⁻¹]	[years]	[\$ ha ⁻¹]	[\$ ha ⁻¹]	[\$ ha ⁻¹]	[\$ ha ⁻¹]
10	857.4	1159.6	24.34	2258.1	2420.0	3544.9	4895.9
11	928.7	1375.1	24.07	2432.6	2586.8	4029.4	5046.7
12	997.7	1572.6	23.80	2590.1	2736.6	4467.9	5222.3
13	1064.3	1749.7	23.56	2728.1	2867.3	4856.4	5407.0
14	1128.7	1905.9	23.32	2846.2	2978.2	5194.4	5588.2
15	1190.9	2041.8	23.10	2944.9	3069.9	5483.6	5757.4
16	1250.9	2158.5	22.90	3025.4	3143.6	5727.0	5909.7
17	1308.8	2257.7	22.70	3089.3	3200.8	5928.2	6042.7
18	1364.6	2341.1	22.52	3138.2	3243.3	6091.0	6155.9
19	1418.4	2410.3	22.35	3173.9	3272.6	6219.4	6249.8
20	1470.3	F 2467.1	22.18	3197.9	3290.5	6317.1	6325.7
21	1520.2	2513.0	22.03	3211.8	I 3298.5	6387.6	6385.0
22	1568.3	2549.4	N 21.88	N 3217.0	3297.9	6434.1	**6429.4
23	1614.6	2577.5	21.74	3214.8	3290.0	6459.5	6460.8
24	1659.0	2598.6	21.61	3206.3	3276.0	J 6466.4	6480.6
25	1701.8	2613.5	21.48	3192.5	3256.8	6457.2	6490.5
26	1742.8	2623.3	21.36	3174.2	3233.3	6434.0	**6491.7
27	1782.2	2628.7	21.25	3152.2	3206.3	6398.7	6485.6
28	1820.0	H 2630.4	21.13	3127.2	3176.4	6352.8	6473.4
29	1856.2	2628.9	21.02	3099.8	3144.2	6298.0	6455.8
30	1890.9	2624.9	20.92	3070.4	3110.2	6235.4	6434.0

$r = 0.06$ p.a.; $p = \$15 \text{ m}^3$; $c_k = \$400 \text{ ha}^{-1}$; $m = 300$; $s = 80$; $b_0 = 100$; $b_1 = 65$.

institutions with collective property rights is effected via multilateral exchange of agreements to take or omit some specified action. Property rights cannot be exchanged within political institutions since the right of alienation (even of shares) is not held by a single person exclusively. The decision to exchange collective property rights via markets can only be reached collectively via political institutions.

Furthermore, within the game theoretical setting, no forest owner has an incentive to cut his tree either earlier, in order to provide additional benefits from thinnings for the other owner(s), or later, in order to provide the public good for the other owner(s). In either case, the expected payoffs from deviating unilaterally from the Nash equilibrium are lower since the other forest owner(s) have incentives to defect. The best response to an early (late) harvest is to harvest late (early). This becomes particularly apparent if $-k$ is assumed to be a large number such that the incentives to free ride on the contributions of others increase. As a consequence, the expected outcomes coincide with the Nash equilibria depicted in Fig. 3.

4.4. Intertemporal coordination

The analysis of the trading equilibria within political institutions has revealed various trading opportunities lacking enforcement mechanisms. With private property rights, trading opportunities are exhausted via market institutions through the exchange of property rights. For political institutions, on the other hand, multilateral trade of agreements is not realizable without additionally introduced constitutional rules (e.g. cost/benefit sharing systems). Depending on the problem situation, there are more or less incentives for the unilateral provision of collectively held goods or services.

The coordination problem, however, changes with the identification of separable property rights units. The analysis above has been restricted to inseparable ownership of single trees (or stands, respectively). If each owner exercises control over several trees (i.e. separable stands) whose value is mutually dependent on the trees of the other owner(s), coordination incentives arise. Consider the corresponding maximization problem

$$\max_{T_i > 0} W^i = \sum_{j \in M_i} [v^{ij}(T_i)e^{-rT_{ij}} - c_j] + w(T_k, T_{-k}) \quad (33)$$

where $M_i = \{1, \dots, m\}$ is the number of trees owned by individual i and $T_i = (T_{i1}, \dots, T_{im})$ are the corresponding harvest ages of the trees. The first-order condition demands that

$$\frac{\partial W^i(T_i^*, T_{-i})}{\partial T_{ih}} = v_{T_{ih}}^{ih} e^{-rT_{ih}} - r v_{T_{ih}}^{ih} e^{-rT_{ih}} + \sum_{j \in M_i, h} v_{T_{ih}}^{ij} e^{-rT_{ij}} + w_{T_{ih}}(T_k, T_{-k}) = 0 \quad (34)$$

$\forall h \in M_i$ and where T_i^* denotes the optimal harvest ages.

Without (recognition of the) mutual dependence between the trees, each forest owner chooses a management regime in accordance with the Hartman harvest ages. Due to the internalized competition effects of the trees owned by a single person, the trees will be cut at different ages (unless there are sufficiently positive inter-tree dependencies).

With relevant external effects between the forest owners, the opportunity arises that each owner cuts some trees later in order to allow for a more efficient public good provision.¹⁹ For instance, if each owner harvests some of his trees at the later harvest age of 26 years in column 8 of Table 2, both have the opportunity to increase their land values by harvesting the rest of the trees at an age of 22 years.

With the definition of a forest management vector, $T_i = (T_{i1}, \dots, T_{im})$, though, defined by the set of harvest ages of each owner, it becomes evident that the coordination problem changes only slightly. If the best response correspondences depicted in Fig. 3 (or Fig. 1, respectively) are constructed on the basis of a management vector, the characteristics of the unique Nash equilibrium remain valid. Each land owner can increase his land value by not contributing to the collective production. If the other owners are delaying the harvest of some of their trees in order to contribute to the public good (or are thinning their stands in order to provide additional revenues for both owners), the forest owner has incentives to harvest all of his trees at an earlier age (or at a later age in the thinning case) which yields a higher individual land value (cf. column 6 (I) in Table 2 for an example).

¹⁹ Of course, the same implicit coordination is possible for the private good production with (non-internalized) external effects such that the exchange of property rights is not necessary, or even inefficient in the face of transaction cost for the transfer of property rights.

The contribution to a higher total and shared land value is thus not self-enforcing. As a consequence, the expected harvest ages tend to be closer together, or even to be equal, compared to a fully internalized situation. With separable investment units, however, and in contrast to the inseparable one-tree (-stand) case considered above, a coordination opportunity emerges in the first place. Without anything to exchange for the promise to contribute, coordination of forest management is impossible altogether.

Coordination opportunities, on the other hand, can become efficient, and thus self-enforceable, by repeated interaction. Consider the case where the stage game G is played repeatedly. Let G^∞ be an extensive game with perfect information and simultaneous moves played infinitely, then $G^\infty = N, H, P, (U_i^\infty)$. Next to the notations of the stage game, H denotes the set of the terminal histories h^t of the action profiles in G such that $h^t = (T^0, T^1, \dots, T^{t-1})$ with $T^j = (T_k^j, T_{-k}^j)$ for each period $j \in H$. The player function $P(h) = N$ assigns the set of all players to every proper sub-history of every terminal history. The payoff space in the infinite game (U_i^∞) defines a discounted payoff to each player according to²⁰

$$W^{i\infty} = \sum_{t=0}^{\infty} \left\{ \sum_{j \in M_i} [v^{ij}(T_i) e^{-rT_{ij}} - c_j] + w(T_k, T_{-k}) \right\} e^{-rt}. \quad (35)$$

In contrast to the one-shot game, each forest owner has the opportunity to react to actions taken by the other owner(s) in previous periods within the repeated game setting.

The equilibria of the repeated game setting depend on the strategies the players can choose. A strategy $s^i(h)$ is an action sequence of a player for every possible history of the repeated game. In order to account for the dynamic nature of the repeated game, subgame perfect equilibria are analyzed which induce Nash equilibria in every subgame of the original game thus accounting for credible threats of punishment. Since the game considered here is infinitely repeated, but at the same time self-similar, each subgame is identical to the whole game. Equilibria can thus be analyzed on the basis of a one-shot deviation principle, which represents a dynamic programming application within the game theoretical framework.

By construction, the strategy to repeatedly play the Nash equilibrium of the stage game is a subgame perfect equilibrium. In each period, the Nash harvest age is the best response to the Nash harvest age chosen previously. Hence, one possible solution pattern for the problem situation faced by forest management is continual repetition of the non-cooperative harvest ages, T^N or T^P .

In contrast to finite games without endogenously affected residual values (as, for example, land values), a Nash equilibrium path is only one of various self-enforcing strategies. Consider, for instance, a grim trigger strategy in which the cooperative strategy is played at all times except if the other forest owner deviates in the preceding period. In this case, the forest owner defects ever after in order to punish the deviation of the other owner. The punishment involves the selection of a “punishment” harvest age according to a minmax strategy defined as

$$T_i^P = \arg \min_{T_{-i} \geq 0} \left\{ \max_{T_i \geq 0} W^i = \sum_{j \in M_i} [v^{ij}(T_i) e^{-rT_{ij}} - c_j] + w(T_k, T_{-k}) \right\}. \quad (36)$$

Depending on the functional relationships, the punishment might, for instance, involve a very early harvest of the own trees in order to reduce the value of the public good.²¹

In this special forestry application, however, another punishment strategy arises based on the interdependence of consecutive periods. If

the punisher refrains from harvesting his trees, the regeneration possibilities of the other owner within the subsequent period might be impaired or even prevented (or, equivalently, his land value decreases). In contrast to other game settings, the forestry application reveals that the players do not only choose their payoff within each period, but they also decide on the length of the periods. Since trees are growing, renewable resources, the length of each period becomes decisive once there is a multi-period setting. With the punishment defined in Eq. (36), the grim trigger strategy becomes

$$s^i(h^t) = \begin{cases} T_i^c & \text{if } T_{-i}^{t-1} = T_{-i}^c \text{ for } t = 1, \dots, t \\ T_i^p & \text{otherwise.} \end{cases} \quad (37)$$

Accordingly, the cooperative action is played at the start after a zero history and after every history in which the other player has chosen the cooperative harvest ages.²² Otherwise, the other player will be punished grimly for deviating.

In general, the infinite repetition of either of the action profiles of the grim trigger strategy leads to land values of

$$W^{i\infty} = \sum_{t=0}^{\infty} \left\{ \sum_{j \in M_i} [v^{ij}(T_i) e^{-rT_{ij}} - c_j] + w(T_k, T_{-k}) \right\} e^{-rt} \\ = \frac{v^i(T_i) e^{rT_i} - c_i + w(T_k, T_{-k})}{1 - e^{-rT_{im}}} \quad (38)$$

for $v^i = \sum_{j=1}^m v^{ij}$, $T_i = (T_{i1}, \dots, T_{im})$ and $t = T_{im}$ since the last harvested tree determines the length of period of the stage game. In this setting, the Faustmann/Hartman rotation ages and land values emerge as an infinitely repeated game with constant strategies of cooperative, homogeneous harvest ages for all trees and players involved.²³

The incentives to deviate are determined by the payoffs associated with the strategies. According to the one-shot deviation principle, and starting with the cooperative harvest ages, deviation is unprofitable if

$$\frac{v^i(T_i^*) e^{rT_i^*} - c_i + w(T_k^*, T_{-k}^*)}{1 - e^{-rT_{im}^*}} \\ \geq v^i(T_i) e^{rT_i} - c_i + w[T_i, b^{-i}(T_i)] + \frac{v^i(T_i^p) e^{rT_i^p} - c_i + w(T_k^p, T_{-k}^p)}{1 - e^{-rT_{im}^p}} e^{rT_i} \quad (39)$$

or

$$v^i(T_i) e^{rT_i} + w[T_i, b^{-i}(T_i)] - v^i(T_i^*) e^{rT_i^*} - w(T_k^*, T_{-k}^*) \\ \leq \frac{v^i(T_i^*) e^{rT_i^*} - c_i + w(T_k^*, T_{-k}^*)}{1 - e^{-rT_{im}^*}} e^{rT_{im}^*} - \frac{v^i(T_i^p) e^{rT_i^p} - c_i + w(T_k^p, T_{-k}^p)}{1 - e^{-rT_{im}^p}} e^{rT_i} \quad (40)$$

i.e. as long as the gain from the one-period deviation on the left hand side is less than the long term gain from cooperation on the right hand side.

In contrast to temporal games, however, in which the length of the stage game period is exogenous, comparisons between the favorability of strategy profiles are more complicated due to potentially diverging lengths of the stage game periods. As a consequence, simple thresholds for discount factors inducing cooperative or non-cooperative outcomes cannot be derived.

²⁰ W^i in the curly brackets can be exchanged for $V^i(T_i, T_{-i})$ for an analysis of the external effect setting.

²¹ The grim trigger strategy does not account for the negative impact of the punishment on the punisher. The punisher might not have incentives to punish forever or even once if the punishment bears high cost for him.

²² With inseparable property rights units, a cooperative strategy can involve the selection of alternating harvest ages for each player.

²³ Furthermore, the Faustmann/Hartman outcomes do only evolve in the absence of asynchronous rotation periods of the corresponding forest owners. Otherwise, the generalized Faustmann outcome emerges (Chang, 1998).

5. Discussion

5.1. Private property rights

As long as property rights are privately and exclusively held, market institutions guarantee the prediction of an unambiguous outcome of the interaction process via bilateral trade. The transaction is efficient, and thus self-enforcing, because not only the trading partners agree on the property rights relevant to them; but at the same time, every bilateral trade implicitly meets the agreement of all other member of the relevant society in terms of agreement on the claims to the traded property rights.

The situation of rights exclusively held by single individuals, however, is not ubiquitous. More often than not, property rights to trees are not exclusively private. In the case of the external effects (as caused by competition between the trees in the example), each owner does not hold the right to determine the value of his trees exclusively since he does not hold the right to urge the other owner to harvest his trees. With the exclusive right to define the harvest ages of his trees, the other owner holds some right to co-determine the value of the other trees (in terms of timber growth).

Private property rights, however, can be exchanged bilaterally via market institutions. An affected owner can buy the trees owned by the other person. If the Nash equilibrium is inefficient relative to transaction cost for the transfer of property rights, trade must take place as long as the forest owners are unwilling or unable to finance the forest out of reserve assets. Alternatively, or in the presence of sufficiently high transaction cost, external effects are accounted for by implicit coordination of forest management as long as the underlying conditions for repeated interactions are satisfied.

Larger forest plots under single ownership, however, often allow for effective forest management at all. The larger a private forest owned by a single individual, the lower are the competition effects to neighboring forest owners relative to the total forest area. With many trees and private owners on a comparatively small area, a coordinated forest management is only realizable if negotiation costs are absent. For forests with low (or negative) land values, on the other hand, transaction or negotiation costs may be too high in relation to the benefits of a transferred property rights set.

The Faustmann rotation age with fully internalized dependencies is restricted to cases in which all non-internalized competition problems have been solved already and completely. If observable private forests differ from the predictions generated by the Faustmann model, the problem analysis must focus either on recent external shifts in the market conditions which have not been internalized yet, or on external effects to the production which are too costly to be internalized via market institutions or unable to be solved by implicit coordination. Depending on the obstacles to exchanges, the Faustmann rotation age might deviate significantly from an efficient Nash equilibrium, or be fully met despite considerable interdependencies between the tree values or external effects.

5.2. Collective property rights

Collective property rights, on the other hand, may cause similar problem situations to the effect that one owner is affected by the actions of another. In contrast to private property rights, however, individual rights to collective property cannot be traded via market institutions. Though collective rights can be exchanged via markets, either with other groups or with single persons, the decision to employ market institutions must be reached collectively. Multilateral agreement is necessary to trade (within) collective ownership. The right to dispose of the collective resource cannot be bought, but can only be transferred by consent of all rights holders.

The inclusiveness of multilateral agreement eliminates potential alternatives for the individual owners and narrows the range within

which the terms of trade can settle (Buchanan, 1975, p. 48). With many participants, it is too costly, in relation to potential gains, to obtain agreement from all. The transaction costs for the transfer of collective property rights through multilateral agreement, as contrasted to those arising from bilateral trade, thus tend to prevent the adaptation to external effects.²⁴

With collective property rights, everybody of the relevant group profits from an individual contribution, or the elimination of a negative external effect, since each holds a right to the public good. The potential dilemma arising within this setting stems from the inability to trade within political institutions. If trees constitute a public good such that all own a right to their harvest, the open access outcome (paragraph 4.2) emerges since the group of all and no one coincides and trees harvested cannot be harvested by another owner of the collective resource. Each contribution of a right holder to a higher land value through delaying the harvest can be enjoyed by other right holders, but cannot be traded.

In order to prolong all harvest ages well beyond the cost-covering diameter, the agreement of all participants is necessary. The individual vote for harvest ages generating a higher quantity/quality of the public good must be met by all others in order to be implemented; it is not possible to simply buy the rights to these harvest ages. Institutions for the (costly) organization of multilateral agreements, such as voting rules, become necessary.²⁵ With high organization cost, there are few incentives to eliminate the negative external effects whereby the outcome after adaptations might deviate substantially from the market outcome.

This conclusion, though, only holds in the absence of conditions favoring implicit coordination. Through repeated interaction, external effects arising out of the collective resource use can basically be reduced. If strategies are adopted which render cooperative outcomes self-enforcing, multilateral agreement is reached implicitly. As long as every participant keeps on contributing to a public good or eliminating a negative external effect, a coordinated solution is viable if actions are repeated. If all abstain from harvesting some of the trees in the open access game, more is left for everybody in subsequent years. Or alternatively, different participants exercising their shared access rights to forests can implicitly agree on allocating the access over the course of the day (e.g. with nocturnally active hunters and diurnally active hikers). Through implicit coordination, new property rights are emerging which solve past interaction conflicts.

5.3. Private and collective property rights

The formulation of the public good problem in the analysis reveals the intermixture of political and market institutions observable in most forests. A forest is typically characterized by different property rights held by different individuals both privately and collectively. For instance, private property rights to trees are “bundled” together with collective rights to access. In this case, and in contrast to market institutions, the Nash equilibrium differs from the individual solution. Compared to the Nash equilibrium in an interaction setting, however, the non-cooperative and the cooperative outcomes (measured in harvest ages) may coincide or deviate in either direction depending on the characteristics of the interdependence.

Analogously, specialization on different products, such as young trees for timber and old trees for recreation, can only be self-enforcing within an infinitely repeated interaction setting (or with a dependent

²⁴ Decision costs increase with the size of the group. In large groups, these cost may require the development of efficient decision rules at the expense of increasing external cost (Buchanan and Tullock, 1962, p. 43ff.).

²⁵ The seemingly potential for consents between the owners to organize resource use differently is imposed by the model construction. With only two persons involved, unanimous consent seems to be readily achievable. With more individuals, the consent is far more complex (and costly) than the employment of social welfare functions implies (Buchanan and Tullock, 1962). The consent to change the rules, however, is more suitably placed at a constitutional stage of economic analysis.

residual value; cf. below). Without enforcement mechanisms, the supply of some old trees falls short of the supply of many (slightly) older trees. Higher land values cannot be achieved through the transfer of individual property rights as within market institutions.

In (most) other situations, though, the groups of rights holders may differ widely. If tree ownership is a private right of some individuals, but the corresponding harvest ages are characterized by collective ownership (as, within limits, can often be observed in harvest regulations), non-tree owners select, i.e. vote for, long (or infinite) harvest ages as long as the present value of the public good increases over rising harvest ages. In the absence of a rule requiring unanimity for the collective decision, tree owners might have to accept harvest ages way beyond the Faustmann age.

In the same way, and based on the same methodological individualism, the analysis of harvest ages in different settings can proceed. For instance, if the management of a forest characterized by publicly (i.e. collectively by every member of the relevant society) held land rights, including the rights to harvest ages and access, has settled on harvest ages generating acceptable potentials for recreational activities, private forest owners, even if interested in recreation (i.e. with private Hartman harvest ages deviating from the Faustmann ages), have no incentives to generate recreation privately for they can use the public forest. The hypothesis might thus be derived that privately provided recreation in forests increases with rising distance from public forests.

As a last example, private forest owners, even if willing to supply nature conservation privately, can be expected to act differently if they share the right to nature conservation in their forest with others collectively. Once the forest owner has chosen harvest ages guaranteeing a specific quantity or quality of nature conservation, the collective right to nature conservation might overrule the private right to the harvest ages. Effectively, the forest owner would lose his private right which offers incentives to prevent nature conservation from being established.

5.4. *De facto property rights*

Property rights matter, though only de facto property rights, i.e., as revealed by human action and as contrasted with formal law (Burton, 2004). A resource officially owned by a collective can effectively be governed by private ownership, and vice versa. An example of this situation, which is frequently analyzed in forest economics, is corruption. Some person is privately selling timber from a public forest. Officially, he is not allowed to unless all members of the relevant society agree to this practice (probably with the help of some decision rule). Put differently, though, the practice implicitly meets the agreement of all members as long as no one objects to it. In this case, the person concerned effectively holds private property rights to the timber through unanimous agreement.

For analyses of conflicts over or in forests arising from the disagreement over the legally enforceable assignment, it is relevant which property rights de facto govern the situation considered. If property rights are officially collective but effectively private, as in the corruption example, the application of political institutions for predictions of the evolution of the forest structure is unwarranted. If corruption practices are effectively governed by private ownership, the analysis must proceed from market institutions. Corruption can be an inefficient land use because the corrupter acts in fear of being expropriated of his effectively private rights at any moment. Resorting to the market theory of bilateral exchanges, it can be hypothesized that the person concerned has any incentives to divest and none to invest such that the forest structure tends to converge to the open access equilibrium.

De facto property rights systems are necessarily consistent while the same must not hold for formal law. Considering a case in which property rights to the harvested trees are collectively owned but the rights to the harvest ages privately held, an unregulated open access outcome can be predicted since the privacy of the harvest ages conflicts with the collectivity of the trees; no participant can be excluded from the

use of the good. The assignment of collective rights to the harvested trees is de facto ineffective as compared with the open access situation. It is thus to be expected that situations which are characterized by collective rights to the use of a resource are often accompanied by additional rules regulating the harvest²⁶ (e.g. by quotas²⁷).

For positive economic theory, it is irrelevant whether the situation considered is to be rated as just or not. What is relevant for the generation of empirically testable hypotheses are the prevailing rules underlying the conflict for they determine the outcome emerging out of the interaction system. The applicability of models is stipulated by the institutions which govern interaction processes, but not by the characteristics or the type of the goods exchanged.

5.5. *The emergence of the Faustmann model*

The analysis has shown that the Faustmann model emerges out of the interaction network under specific conditions. It is bound to fully internalized interdependencies and stationary strategies over time. On the one hand, the Faustmann model thus represents a very special case of infinitely repeated interactions. One-shot deviations from the strategies of any player involved, external effects or de facto collective property rights tend to prevent the Faustmann outcome from evolving. In view of the various opportunities to defect, the Faustmann rotation age appears less likely to be observable.

On the other hand, however, the Faustmann model reveals how impersonal market institutions allow for an intertemporal coordination of forest management by innumerable individuals over virtually infinite time horizons. Guided by the abstract rules of market exchanges, all individuals in every generation cooperate as if maximizing the land value. The Faustmann model as a genuine market model indicates how infinitely repeated, non-cooperative games are restricted to large group settings.

The game theoretical framework tends to disguise large group interactions within the extended order. With personal interactions within non-anonymous groups, strategic behavior on the part of the members offers incentives to defect thus preventing the Faustmann outcome from emerging. In large anonymous groups, by contrast, strategic behavior is ineffective as the actions of other members cannot be influenced. The great defection potential in the repeated game of forest management, as revealed by the analysis, disappears in the extended order. The concentration on strategic behavior violates the institutions governing market exchanges in large groups.

The same coordination principle allows for the inclusion of infinite time horizons. Forest owners might not live forever, but market exchanges can. Through the integration of residual values which are influenced by current actions, the outcomes of infinitely repeated games can be described in terms of one-shot games. In economic theory, however, just the reverse relationship is relevant: the outcome of current actions with residual values can be described by infinite time horizons. Forest owners are not required to assess the consequences of their current decisions on any future rotation period, but on the current forest value. The infinite Faustmann model does not describe individual action over time, but rather the outcome of interactions within complex social networks.

6. Conclusions

Property rights matter. They matter not just because they determine the rents accruing from the use and regulation of resources, but because

²⁶ The ability of collective resource owners (in small groups) to overcome the "tragedy of the commons" through additional rules of conduct has repeatedly been stressed (e.g. Ostrom, 2010).

²⁷ Individual transferable quotas have come into fashion for solving the open access dilemma in ocean fishery (Conrad, 2010, p. 76). Effectively, the quotas represent the transformation of the open access resource to a collective resource. A consequence of this process can be the exclusion of former or potential future users of the resource, which might constitute an obstacle to the implementation process.

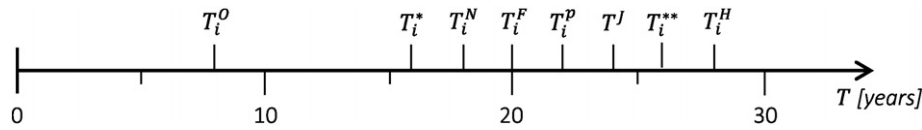


Fig. 4. Compilation of the harvest ages from the examples in Tables 1 and 2.

they represent a consent within the complex systems of modern societies. The way property rights are distributed among individuals through mutual recognition defines the expected outcome of the exchange process.

The analysis has shown that predictions of objectively observable harvest and regeneration ages based on forest economic theory can vary considerably depending on the property rights assignment and the ensuing institutional setting for exchanges. Different silvicultural systems and practices, as characterized by diverging harvest ages of trees, are the consequence of different institutional settings. The management of trees cannot be judged independent of the institutions governing human interactions. With diverse institutional structures, single silvicultural solutions are not viable; silvicultural diversity emerges with institutional diversity.

Privately optimal harvest ages, as implied by the Faustmann or Hartman models, are restricted to fully internalized interdependence settings. Once the rights to forests are not entirely recognized as private rights, different forest structures evolve. Politically coordinated forest management differs from privately held rights to the effect that external effects not coordinated implicitly cannot be internalized via exchanges of individual property rights.

Fig. 4 summarizes the range of the different harvest ages from the examples in Tables 1 and 2. All harvest ages displayed are optimal or efficient. The optimality or efficiency of forest management regimes cannot be judged independently of the set of rules governing the problem situation, but hinges entirely on the institutional arrangements that evolved out of past interactions. With it, universal standards for the determination of efficient, self-enforcing or sustainable states or developments of forests become untenable.

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Appendix A

The functions to the examples have been selected because of their frequent employment for illustrations in the forest economic literature. Timber growth function:

$$q^i(T_i, T_{-i}) = \exp\left(9.75 - \frac{3,418.11}{m * T_i} - \frac{740.82}{T_i * s} - \frac{34.01}{T_i^2} - \frac{1,527.67}{s^2}\right) 0.99914^{T_i T_{-i}} \quad (A.1)$$

where m denotes the initial density and s the site index. Present value of public good:

$$w(T_i, T_{-i}) = b_0(T_i + T_{-i})e^{-(1/b_1)(T_i + T_{-i})} \quad (A.2)$$

where b_0 and b_1 are constants.

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